

The Neutrino Manual

A small functional array language

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This manual covers Neutrino as implemented: every example below was executed against the interpreter and shows its actual output. For a quick pitch and build instructions see the [README](#); for the honest list of sharp edges see [KNOWN_LIMITATIONS](#).

1. Getting started

Build (a C23 compiler and libm; readline optional — see the README for macOS notes):

```
make          # ./neutrino, the REPL
make test     # golden suite + codegen goldens
```

Start the REPL and type an expression; its value echoes back:

```
neutrino> 2 + 3 * 4
14
neutrino> [1, 2; 3, 4] * [5; 6]
[ 17
 39 ]
```

A trailing `;` evaluates a statement but suppresses the echo. `#` and `%` both start a comment that runs to end of line. Ctrl-D exits; Ctrl-C cancels the current input.

2. The REPL

The interactive shell adds conveniences on top of the language. None of them apply to scripts — they are REPL features.

Commands (typed bare, not as function calls):

Command	Effect
help / help(f)	catalogue of all builtins / details on one
who	your variables, with type and shape
format ...	number display: format long, format short e, format(8), format to show
pretty on\ off	aligned multi-line matrix display (default on in the REPL)
more on\ off	page long output through \$PAGER (default off)
!cmd	run a shell command (!ls, !git status)
dis(f)	disassemble a function's bytecode

Autocall. A bare name that is a zero-argument builtin or closure is called: who, help, rand work without parentheses, and so does your own `let f = fn -> 42; f`.

Line editing. With readline (or macOS libedit), you get history (persisted to `~/.neutrino_history`), completion on builtin and variable names, and the usual Emacs keys. Multi-line constructs continue with a `...>` prompt until the end arrives.

Display defaults. The REPL prints matrices as aligned blocks (`pretty on`) and numbers in a terse 6-significant-digit format; both are configurable — see [Output and formatting](#).

3. Values and types

Neutrino has nine value kinds. The scalar kinds:

Type	Literals	Notes
Int	42, -7	64-bit signed; overflow wraps silently (documented footgun)
Float	3.14, 1e-9, 2.5e3	IEEE double
Bool	true, false	distinct from numbers: <code>1 == true</code> is an error
Complex	2i, 1 + 3i, 2.5i	double re/im pair
String	"hello"	inert: strings display, print, and pass around, but have no operations — no concatenation, comparison, or indexing (yet)
Null	null	the “no value” value; a suppressed or valueless statement yields it

And the compound kinds: Array (the 2-D numeric matrix — every array is rows x cols; a scalar is *not* a 1x1 array), Record (`{x = 1, y = 2}`, fields via `.x`), and Function (builtins and closures).

The type discipline is strict rather than coercive: Bool does not silently become a number in arithmetic or comparison (`1 == true` errors), and mixing Bool with numerics in an array literal

is an error. The one deliberate blur: Int and Float compare and combine freely (`5 == 5.0` is true), and `/` always produces a Float (`4 / 2` is `2.0`).

Floating point behaves like floating point:

```
neutrino> 0.1 + 0.2 == 0.3
false
neutrino> 5 / 0
inf
```

Division by zero yields `inf/nan` rather than an error — test with `isfinite/isnan` where it matters.

4. Operators and expressions

From loosest to tightest binding:

Level	Operators	Notes
pipe	<code>\ ></code>	left-assoc; see Functions
logical or	<code>\ \ </code>	short-circuit
logical and	<code>&&</code>	short-circuit
elementwise or	<code>\ </code>	on logicals/arrays
elementwise and	<code>&</code>	on logicals/arrays
comparison	<code>== ~= < <= > >=</code>	elementwise on arrays -> logical array
range	<code>a:b, a:step:b</code>	inclusive; float steps fine
additive	<code>+ -</code>	
multiplicative	<code>* / .* ./ \ \.</code>	<code>*</code> is the matrix product on matrices; <code>.*</code> elementwise; <code>\</code> left division (solve)
unary	<code>- ! ~</code>	binds looser than <code>^</code> : <code>-2^2</code> is <code>-4</code>
power	<code>^ .^</code>	right-assoc: <code>2^3^2</code> is 512; <code>^</code> on a square matrix is the matrix power
postfix	<code>' .' f(x) a[i] .field</code>	<code>'</code> is conjugate transpose, <code>.'</code> plain

The `.-`-prefixed operators are the elementwise family, exactly as in Octave:

```
neutrino> [1, 2, 3] .* [4, 5, 6]
[4, 10, 18]
neutrino> 2 .^ [1, 2, 3]
[2, 4, 8]
neutrino> [1i, 2]'          # conjugate transpose
[ -1i
  2+0i ]
```

Scalars broadcast against arrays in every elementwise operation (`[1, 2] + 1` is `[2, 3]`); two arrays must agree in shape exactly.

5. Variables and scope

`let name = value` is the binding **statement**: at the top level it creates a global; inside a loop body or block it creates a local scoped to that construct. A bare `name = value` (no `let`) is

assignment: it walks outward through the enclosing scopes and updates the nearest existing name — this is how a loop body updates an accumulator defined outside it.

`let name = value in body` is the binding **expression**: `name` is visible only in `body`, and the whole thing evaluates to `body`. Bindings nest and shadow:

```
neutrino> let a = 1 in a + (let a = 100 in a) + a
102
```

6. Control flow

`if` is an expression:

```
neutrino> let x = 5; if x > 3 then "big" else "small" end
"big"
```

The `else` branch is optional; if the condition is false and there is no `else`, the result is `null`. Conditions must be `Bool` — a number is not a condition.

Loops are statements (they yield `null`):

```
neutrino> let s = 0; for i = 1:10 do s = s + i end; s
55
neutrino> let n = 1; while n < 100 do n = n * 2 end; n
128
```

`break` leaves the nearest loop, `continue` skips to its next iteration, and `return [value]` exits the enclosing function. All three are safe anywhere — including mid-expression (`acc + (if bad then continue else v end)`): the VM restores the loop's stack state on a non-local exit.

Block expressions sequence statements inside parentheses; the value is the final expression, and `let` bindings are local to the block:

```
neutrino> (let x = 3; let y = 4; sqrt(x*x + y*y))
5
neutrino> (let q = 7; q); q
error: undefined name 'q'           # block locals do not leak
```

This is the natural shape for a multi-step function body.

7. Functions

`fn params -> expr` is a lambda; its body is one expression (use a block expression or `let ... in` for multiple steps). Functions are values: bind them, pass them, return them.

```
neutrino> let f = fn x -> x ^ 2 + 1; f(3)
10
neutrino> let add = fn a -> fn b -> a + b; add(2)(5)    # currying
7
neutrino> let g = fn x -> (let s = x * x; s + 1); g(4)
17
```

Closures capture by value at creation. Recursion works through the function's own name.

Sections. A parenthesised expression containing `_` becomes a lambda; each `_` is a fresh parameter, left to right: `(_ + 1)` is `fn x -> x + 1`, `(_ * _)` takes two arguments.

map. `map(f, A)` applies `f` elementwise: `map((_ * 10), [1, 2, 3])` is `[10, 20, 30]`.

The pipe. `x |> rhs` feeds a value forward. If `rhs` mentions `@`, the pipe binds `@` to `x` and evaluates `rhs`; if `rhs` is a bare callable, the pipe applies it:

```

neutrino> [1, 2, 3, 4] |> sum(@) |> sqrt
3.16228
neutrino> 5 |> @ + 1
6
neutrino> 9 |> sqrt                # bare callable: sqrt(9)
3

```

Pipes chain left to right, which reads as a data-flow pipeline.

8. Arrays and matrices

Matrix literals use `,` between columns and `;` between rows; every array is two-dimensional and **1-indexed**. `[]` is the empty array.

Indexing supports scalars, ranges, `:` (everything along a dimension), `end` (the last index, with arithmetic), index vectors (`gather`), and logical masks:

```

neutrino> let A = [1, 2, 3; 4, 5, 6]
[ 1  2  3
  4  5  6 ]
neutrino> size(A)
[2, 3]
neutrino> A[2, 3]          # element
6
neutrino> A[1, : ]        # row
[1, 2, 3]
neutrino> A[:, 2]         # column
[ 2
  5 ]
neutrino> A[end, end]
6
neutrino> let v = [10, 20, 30, 40]; v[2:3]
[20, 30]
neutrino> v[v > 15]        # logical mask
[20, 30, 40]
neutrino> v[[1, 4]]        # gather
[10, 40]

```

Indexed assignment works with the same selectors, copy-on-write: `A[1, 2] = 9`, `v[v < 0] = 0`, `A[2, :] = [7, 8, 9]`. The target must be a plain name and indices must be in range (no auto-growing).

`unique(A)` returns the sorted distinct elements — vectors keep their orientation, matrices flatten to a row. **Logical arrays** come from comparisons and drive masking and counting: `sum(A > 2)` counts, `any/all` test, `find(mask)` gives 1-based positions, `where(mask, a, b)` selects elementwise.

Construction and reshaping: `zeros`, `ones`, `eye`, `diag`, `linspace`, `reshape` (row-major), `repmat`, `ranges 1:n`. **Reductions** take an optional dimension: `sum(A, 1)` down columns, `sum(A, 2)` across rows, `max(A, [], dim)` for the extrema.

Descriptive statistics follow the same conventions. `var` and `std` normalize by `N-1` by default (`var(A, 1)` divides by `N`), `median` handles even counts by averaging, and `quantile` uses linear interpolation between order statistics (the NumPy default):

```

neutrino> let x = [2, 7, 4, 9, 3]
neutrino> var(x)
8.5

```

```
neutrino> quantile(x, [0.25, 0.5, 0.75])
[3, 4, 7]
```

`cov(X)` and `corr(X)` treat a matrix's **columns as variables** and rows as observations, returning the $p \times p$ covariance / Pearson correlation matrix (`cov` takes the same w normalization as `var`). On two vectors they return the scalar: `cov(x, y)`, `corr(x, y)`. A constant column has no correlation to speak of — those entries are `nan`:

```
neutrino> corr([1, 2, 3, 4, 5], [2, 1, 4, 3, 5])
0.8
```

9. Linear algebra

`*` is the matrix product; `\` solves. Square systems use LU with partial pivoting; non-square `\` is least squares — overdetermined gives the LS fit, underdetermined the minimum-norm solution.

```
neutrino> [2, 1; 1, 3] \ [3; 5]
[ 0.8
  1.4 ]
neutrino> [1, 1; 1, 2; 1, 3] \ [1; 2; 2]      # regression: intercept, slope
[ 0.666667
  0.5 ]
```

The decompositions return records, so the pieces have names:

Call	Returns	Identity
<code>lu(A)</code>	<code>{L, U, p}</code>	$P*A = L*U$
<code>qr(A)</code>	<code>{Q, R}</code>	$A = Q*R$
<code>chol(A)</code>	<code>L</code>	$L*L' = A$ (Hermitian PD)
<code>svd(A)</code>	<code>{U, S, V}</code>	$A = U*diag(S)*V'$
<code>eig(A)</code>	<code>{values, vectors}</code>	$A*V = V*diag(values)$

`eig` handles both Hermitian matrices (Jacobi; real ascending eigenvalues, orthonormal vectors) and general ones (complex QR + inverse iteration; complex pairs come out right). All of it is complex-capable — `'` being the conjugate transpose is what makes the identities hold. `kron(A, B)` is the Kronecker product, `det`, `inv`, `trace`, `norm`, `dot` round out the set.

10. Complex numbers

The imaginary literal is a number followed by `i`. Complex values propagate through arithmetic and the math library; results stay complex (`1i * 1i` is `-1+0i`, not `-1`). `real`, `imag`, `conj`, `angle` access the parts, `abs` is the modulus. Functions with limited real domains return complex off it: `sqrt(-4)` is `2i`, `log(-1)` is `3.14159i`, `asin(2)` is complex.

11. Special functions and statistics

The special-function set is chosen as *primitives* from which the classical distributions are one-liners:

```
neutrino> let normcdf = fn x -> 0.5 * erfc(-x / sqrt(2))
neutrino> normcdf(1.96)
0.975002
neutrino> norminv(0.975)
1.95996
```

`gammainc(x, a)` is the regularized lower incomplete gamma — the chi-square CDF is `gammainc(x/2, k/2)`. `betainc(x, a, b)` is the regularized incomplete beta — Student-t and F CDFs fall out of it. `erf/erfc`, `beta/lbeta`, `gamma/lgamma`, `digamma`, and integer-order Bessel `besselj/bessely` complete the set. All are real-domain, elementwise over arrays, and validated against SciPy.

Root finding, minimization, integration

These three take a **function argument** — a closure or builtin — and call it repeatedly. `fzero(f, a, b)` finds a root by Brent's method (`f(a)` and `f(b)` must differ in sign); `fminbnd(f, a, b)` minimizes on an interval and returns `{x, fx}`; `integral(f, a, b[, tol])` integrates adaptively (Simpson with Richardson estimate; finite limits; default tolerance `1e-10`). Closures capture data, so parameterized problems read naturally — a bond's yield to maturity:

```
neutrino> let cf = [100, 100, 100, 1100]
neutrino> let npv = fn r -> sum(cf ./ ((1 + r) .^ (1:4))) - 1000
neutrino> format(6); fzero(npv, 0.01, 0.3)
0.100000
```

A divergent or wildly oscillatory integrand fails with an error rather than a wrong answer; infinite limits are rejected — substitute to a finite domain first.

12. Random numbers

The generator is `xoshiro256**` seeded through `splitmix64`, and it is **reproducible by default**: every fresh session starts from the same fixed seed, so a script's random draws are stable run to run. `rng(seed)` reseeds — same seed, same stream. `rand`, `randn`, `randi` draw uniform, normal, and integer variates, scalar or matrix-shaped (`rand(3)`, `randn(2, 4)`).

13. Plotting

Plotting is delegated to **gnuplot**, out of process — a soft dependency: the language works without it, and `plot` reports cleanly if it is missing (`plot: gnuplot failed (exit 127) – is gnuplot installed?`).

`plot(y)` plots a vector against its index; `plot(x, y)` plots pairs; if `y` is a **matrix**, each column is a separate series. An optional trailing argument is either a gnuplot style string ("`points`", "`lines lw 2`", "`impulses`") or an options record:

```
neutrino> let x = linspace(0, 10, 200)
neutrino> plot(x, map(sin, x), {title = "sin(x)", xlabel = "x", grid = true})
```

Recognised options: `title`, `xlabel`, `ylabel`, `style` (strings); `logx`, `logy`, `grid` (booleans); `xrange`, `yrange` (`[lo, hi]` vectors) to fix an axis instead of letting gnuplot choose; and legend labels — `label` for a single series, `label1`, `label2`, ... for several (unlabeled series keep the series `k` default):

```
neutrino> let t = (linspace(0, 6.28, 100))';
neutrino> plot(t, [map(sin, t), map(cos, t)], {label1 = "sin", label2 = "cos"})
``` `hist(y)` draws a histogram
(`hist(y, nbins)` to choose the bin count; the default follows Sturges' rule),
and accepts the same trailing options record:
neutrino> rng(7) neutrino> hist(randn(1, 5000), 40)
```

A word of caution that ``yrange`` exists to address: gnuplot auto-ranges the y-axis to the data, which can make pure sampling noise look like structure. A

histogram of 100 000 uniform draws in 20 bins has bin counts of 5000 +/- 69 (one binomial standard deviation) — auto-ranged, that +/-1.4% wiggle fills the whole plot and looks alarming; anchored at zero it is the flat wall it should be:

```
neutrino> hist(rand(1, 100000), 20, {yrange = [0, 6000]})
```

Plots open in a gnuplot window that outlives the command (``gnuplot -persist``). For scripted rendering, two environment variables redirect output — ``NEUTRINO_PLOT_TERM`` sets the gnuplot terminal and ``NEUTRINO_PLOT_OUT`` the file:

```
```sh
NEUTRINO_PLOT_TERM="pngcairo size 800,500" NEUTRINO_PLOT_OUT=fig.png \
  ./neutrino script.nu
```

(`NEUTRINO_PLOT_TERM="dumb size 76,20"` draws ASCII plots straight into the terminal, which is occasionally exactly what you want.) Complex data is rejected — plot `real(z)`, `imag(z)`, or `abs(z)` explicitly.

14. Data files

`readcsv(file)` reads a numeric CSV into a Float matrix; `writcsv(file, A)` writes one at full precision, so values **round-trip bit-exactly**. Empty cells become `nan` (missing data), Windows line endings are tolerated, and `{delim = ";", skip = n}` options handle other separators and preamble lines. Ragged rows and non-numeric cells are errors that name the row and column.

`readtable(file)` reads a CSV whose first line is a header and returns a **record of column vectors**, keys sanitized from the column names ("GDP Growth (%) becomes `gdp_growth`; duplicates get `_2`, `_3`, ...). This is Neutrino's data frame — named columns plus the existing mask machinery:

```
neutrino> writcsv("/tmp/m.csv", [1, 2; 3, 4]); readcsv("/tmp/m.csv")
[ 1  2
  3  4 ]

neutrino> let d = readtable("tests/data/macro.csv")
neutrino> mean(d.gdp_growth)
1.93333
neutrino> d.cpi[d.year >= 2021]
[ nan
   8 ]
```

For a headerless file use `readcsv` — `readtable` will take the first line as a header regardless of its content. A column containing text (country names, tickers) is rejected with an error naming the column: representing it needs first-class strings, which the language does not yet have.

15. Output and formatting

`format` controls how numbers print. Explicitly chosen formats keep trailing zeros for consistent width; the startup default is terse:

```
neutrino> format(4)
neutrino> for i = 1:3 do print(sqrt(i)) end
1.000
1.414
1.732
```


`print` takes either plain values (space-separated) or a template whose `{}` holes are filled in order; a hole can carry its own spec `{:[-][width][.prec][f|e|g]}` for per-hole width, precision, and conversion:

```
neutrino> print("sqrt({}) = {:8.4f}", 2, sqrt(2))
sqrt(2) = 1.4142
```

Strings print without quotes in `print`; the REPL echo shows them quoted (the echo is a representation, `print` is output). `pretty on/off` toggles the aligned matrix display; `more` on pages long output.

16. Scripts and tools

`neutrino file.nu` runs a script (top level is a statement sequence; `#!/%` comments). The binary also exposes the compiler pipeline:

Invocation	Shows
<code>neutrino --tokens file.nu</code>	the token stream
<code>neutrino --ast file.nu</code>	the parse tree
<code>neutrino --dis file.nu</code>	the compiled bytecode, statement by statement

`tic` starts a monotonic wall-clock timer and `toc()` returns the elapsed seconds — bare `toc` as a statement echoes it, but in an expression position write `toc()` (a bare name is a value reference, as with any function):

```
neutrino> tic; let A = randn(100, 100); let B = A * A; toc() < 60
true
```

`vmtest` is the headless driver used by the test suite: it reads `stdin` line by line and echoes each result, so `printf 'sum(1:100)\n' | ./vmtest` prints 5050. `dis(f)` disassembles from inside the language. The golden suite (`make test`) and its ASan twin (`make test-asan`) are how changes prove themselves; `tests/dis/` pins the emitted bytecode for core constructs.

17. Builtin reference

Generated from the interpreter's own documentation table (the same data `help` shows), so it cannot drift from the implementation.

Core & introspection

Signature	Description
<code>print(...)</code> <code>print(tmpl, ...)</code>	print values; template fills <code>{}</code> in order; <code>{:[-][w][.p][f}</code>
<code>who</code>	list the variables you have defined (name, type, shape)
<code>help / help(f)</code>	help lists every builtin; <code>help(f)</code> describes one
<code>system(cmd)</code>	run a shell command string; return its exit status
<code>dis(f)</code>	disassemble a function's bytecode (compiler/VM introspection)
<code>format / format(m)</code>	show or set number display: "short", "long", "short e", or a digit count

Signature	Description
<code>size(x)</code>	[rows, cols] of x (a scalar is 1x1)
<code>length(x)</code>	longest dimension of x (0 if empty)
<code>numel(x)</code>	number of elements (rows*cols)
<code>tic</code>	start the wall-clock timer (monotonic)
<code>toc</code>	seconds elapsed since tic

Solvers

Signature	Description
<code>fzero(f, a, b)</code>	root of f in [a, b] (Brent; f(a), f(b) must differ in sign)
<code>fminbnd(f, a, b)</code>	minimum of f on [a, b] (Brent) -> {x, fx}
<code>integral(f, a, b[, tol])</code>	definite integral (adaptive Simpson, finite limits; default tol 1e-10)

Data files

Signature	Description
<code>readcsv(file[, opts])</code>	numeric CSV -> Float matrix; empty cells are nan; opts: {delim, skip}
<code>writcsv(file, A[, opts])</code>	matrix -> CSV, full precision (round-trips); opts: {delim}
<code>readtable(file[, opts])</code>	CSV with a header -> record of column vectors named from the header

Plotting

Signature	Description
<code>plot(y) plot(x, y) plot(x, Y, opts)</code>	line plot via gnuplot; Y columns are series; opts: style string or {title, xlabel, ylabel, style, logx, logy, grid, xrange, yrange, label, label1..labelN}
<code>hist(y[, nbins][, opts])</code>	histogram via gnuplot; opts as in plot (yrange to anchor the axis, label for the legend)

Array construction

Signature	Description
<code>zeros(r, c)</code>	r-by-c matrix of zeros
<code>ones(r, c)</code>	r-by-c matrix of ones
<code>eye(n)</code>	n-by-n identity matrix
<code>diag(x)</code>	vector -> diagonal matrix; matrix -> its diagonal as a column
<code>linspace(a, b, n)</code>	row of n points evenly spaced from a to b inclusive

Signature	Description
<code>reshape(A, r, c)</code>	reinterpret A's elements as r-by-c (row-major), element count must match
<code>repmat(A, m, n)</code>	tile A into an m-by-n grid of copies

Reductions

Signature	Description
<code>sum(A) sum(A, dim)</code>	sum of all elements, or along dim (1 = columns, 2 = rows)
<code>prod(A) prod(A, dim)</code>	product of all elements, or along dim
<code>cov(X[, w]) cov(x, y[, w])</code>	covariance matrix of X's columns (rows = observations), or scalar cov of two vectors; w as in var
<code>corr(X) corr(x, y)</code>	Pearson correlation matrix of X's columns, or scalar correlation of two vectors
<code>var(A) var(A, w) var(A, w, dim)</code>	variance; w = 0 divides by N-1 (default), w = 1 by N
<code>std(A) std(A, w) std(A, w, dim)</code>	standard deviation (sqrt of var, same normalization)
<code>median(A) median(A, dim)</code>	median of all elements, or along dim
<code>quantile(x, p)</code>	quantile(s) of the data at probability p (scalar or vector); linear interpolation
<code>mean(A) mean(A, dim)</code>	mean of all elements, or along dim
<code>min(A) min(a, b) min(A, [], dim)</code>	smallest element; elementwise min; or min along dim
<code>max(A) max(a, b) max(A, [], dim)</code>	largest element; elementwise max; or max along dim
<code>any(mask) any(mask, dim)</code>	true if any element is nonzero/true (overall or along dim)
<code>all(mask) all(mask, dim)</code>	true if every element is nonzero/true (overall or along dim)

Array utilities

Signature	Description
<code>unique(A)</code>	sorted distinct elements; vectors keep orientation, matrices flatten to a row
<code>sort(A)</code>	ascending sort: a vector as a whole, a matrix by column
<code>find(mask)</code>	1-based positions of nonzero/true elements (row-major)
<code>where(mask) where(mask, a, b)</code>	indices of true, or pick a where true and b where false
<code>cumsum(A)</code>	cumulative sum along a vector, or down each column
<code>cumprod(A)</code>	cumulative product along a vector, or down each column
<code>diff(A)</code>	consecutive differences along a vector, or down each column

Signature	Description
<code>flipud(A)</code>	reverse row order (flip up-down)
<code>fliplr(A)</code>	reverse column order (flip left-right)

Mathematical functions

Signature	Description
<code>abs(x)</code>	absolute value, or complex magnitude
<code>sqrt(x)</code>	square root (complex result for negative reals)
<code>cbrt(x)</code>	real cube root
<code>exp(x)</code>	e raised to the x (complex-aware)
<code>log(x)</code>	natural logarithm (complex for negatives)
<code>ln(x)</code>	natural logarithm (alias for log)
<code>log10(x)</code>	base-10 logarithm (complex for negatives)
<code>log2(x)</code>	base-2 logarithm (complex for negatives)
<code>sign(x)</code>	-1 / 0 / +1 by sign; z/
<code>floor(x)</code>	round toward -infinity (componentwise on complex)
<code>ceil(x)</code>	round toward +infinity (componentwise on complex)
<code>round(x)</code>	round to nearest (componentwise on complex)
<code>trunc(x)</code>	round toward zero
<code>hypot(a, b)</code>	$\sqrt{a^2 + b^2}$ without overflow (elementwise)
<code>mod(a, b)</code>	modulo, result takes the sign of b (elementwise)
<code>rem(a, b)</code>	remainder, result takes the sign of a (elementwise)
<code>gamma(x)</code>	gamma function (real, elementwise)
<code>erf(x)</code>	error function (real, elementwise)
<code>erfc(x)</code>	complementary error function $1 - \text{erf}(x)$
<code>beta(a, b)</code>	beta function ($a, b > 0$, elementwise)
<code>lbeta(a, b)</code>	log of the beta function
<code>gammainc(x, a)</code>	regularized lower incomplete gamma $P(a, x)$ (the χ^2 CDF)
<code>betainc(x, a, b)</code>	regularized incomplete beta $I_x(a, b)$ (Student-t / F CDFs)
<code>norminv(p)</code>	standard normal quantile (inverse CDF)
<code>digamma(x)</code>	digamma $\psi(x) = d/dx \log \gamma(x)$
<code>besselj(n, x)</code>	Bessel function of the first kind, integer order n
<code>bessely(n, x)</code>	Bessel function of the second kind, integer order n ($x > 0$)
<code>lgamma(x)</code>	log of

Linear algebra

Signature	Description
kron(A, B)	Kronecker product: (m x n) kron (p x q) -> (mp x nq)
dot(a, b)	inner product of two vectors
norm(x) norm(x, p)	vector p-norm (p = 1 or 2, default 2); matrix Frobenius norm
trace(A)	sum of the diagonal
det(A)	determinant via LU
inv(A)	matrix inverse (solves $A \cdot I$)
lu(A)	LU with partial pivoting -> {L, U, p}, so $PA = LU$
qr(A)	Householder QR -> {Q, R} (real or complex)
chol(A)	Cholesky factor L (lower), $L^*L = A$ (SPD / Hermitian PD)
eig(A)	eigendecomposition -> {values, vectors}; Hermitian (ascending real) or general (complex)
svd(A)	thin SVD -> {U, S, V}, $A = U \text{diag}(S) V'$ (S descending)

Trigonometric & hyperbolic

Signature	Description
sin(x)	sine (complex-aware, elementwise)
cos(x)	cosine (complex-aware, elementwise)
tan(x)	tangent (complex-aware, elementwise)
asin(x)	arcsine (complex outside [-1, 1])
acos(x)	arccosine (complex outside [-1, 1])
atan(x)	arctangent (complex-aware)
atan2(y, x)	two-argument arctangent (elementwise)
sinh(x)	hyperbolic sine (complex-aware)
cosh(x)	hyperbolic cosine (complex-aware)
tanh(x)	hyperbolic tangent (complex-aware)
asinh(x)	inverse hyperbolic sine (complex-aware)
acosh(x)	inverse hyperbolic cosine (complex below 1)
atanh(x)	inverse hyperbolic tangent (complex outside (-1, 1))

Complex accessors

Signature	Description
real(z)	real part (elementwise)
imag(z)	imaginary part (elementwise)
conj(z)	complex conjugate (elementwise)
angle(z)	argument atan2(im, re) (elementwise)
arg(z)	argument atan2(im, re) (alias for angle)

Random numbers

Signature	Description
rng(seed)	reseed the generator (xoshiro256**); same seed, same stream
rand() rand(n) rand(r, c)	uniform draws on [0, 1)
randn() randn(n) randn(r, c)	standard-normal draws
randi(imax[, r, c]) randi([lo, hi], ...)	uniform random integers

Predicates

Signature	Description
isnan(x)	elementwise test for NaN -> logical
isinf(x)	elementwise test for +/-Inf -> logical
isfinite(x)	elementwise test for a finite value -> logical

Higher-order functions

Signature	Description
map(f, A)	apply f to each element of A, returning an array of results

18. Grammar summary

Reserved words: let in fn if then else end true false null for while do break continue return.

```

program    := statement*
statement  := 'let' NAME '=' expr                # binding (global at top level)
            | NAME '=' expr                      # assignment (walks scopes)
            | lvalue '[' indices ']' '=' expr
            | 'for' NAME '=' expr 'do' statement* 'end'
            | 'while' expr 'do' statement* 'end'
            | 'break' | 'continue' | 'return' expr?
            | expr
expr        := 'let' NAME '=' expr 'in' expr
            | 'if' expr 'then' expr ('else' expr)? 'end'
            | 'fn' NAME* '->' expr
            | '(' statement ';' statement)* ')'    # block expression
            | expr binop expr | unop expr | expr postfix
            | NAME | literal | '[' rows ']' | '{' fields '}'
            | expr '|>' expr
postfix     := '"' | "." | '(' args ')' | '[' indices ']' | '.' NAME

```

Statements separate by newline or ; (a trailing ; also suppresses the REPL echo). Comments run from # or % to end of line.

Every example in this manual was executed against the current interpreter; the builtin reference is generated from the source. If you find a discrepancy, that is a bug — please report it.